Introduction To Algorithms

Week 5 – **Recursion, Iteration, and Strategy**

"Same Problem, Different Strategy"

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Terms to Remember

Term	Brief Meaning
Strategy	A general approach for solving a problem
Correctness	Whether the algorithm produces the expected result
Performance	How efficiently the algorithm uses work or resources
Scalability	How the approach behaves as size or demand grows
Elasticity	Ability to handle changing demand or scale when resources change
Polynomial	A growth pattern that can be expressed with powers such as n, n2, or n3
NP-Hard	A class of problems that are at least as hard as the hardest problems in NP
NP-Complete	Problems that are both in NP and as hard as any problem in NP
Divide and conquer	Split, solve smaller pieces, then combine
Dynamic programming	Store and reuse results from overlapping subproblems
Greedy strategy	Make the best-looking choice at each step
Recursion	A function solves a smaller version of the same problem
Base case	The condition where recursion stops

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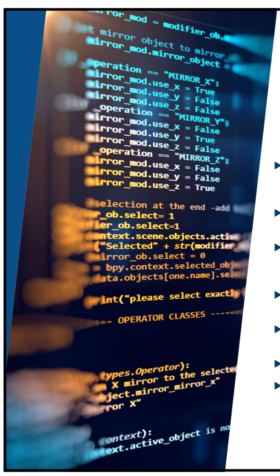

What We Will Use Today

- Correctness, performance, and scalability as strategy questions
- Divide and conquer as a split-and-combine pattern
- Dynamic programming as recognition of saved repeated work
- Greedy as local choice with possible risk
- Iteration and recursion as two strategy forms
- Comparison table evidence
- Strategy recommendation with conditions

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What We Will "Touch On" and Revisit Later

- Graphs and traversal strategy
- Recommendation and ranking systems
- Larger-scale algorithms
- Explainability and limits of algorithmic solutions
- Final-assessment explanation and justification

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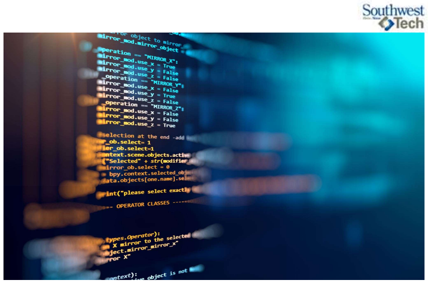

Review Coding Lab

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Review: What Lab 04 Taught Us

Last week, we learned that:

Search strategy depended on assumption

Binary search was powerful only when:

- The data was sorted
- The comparison rule matched the order
- The trace decisions were valid

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Today's Question

How can two correct solutions still be different in quality?

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Same Problem, Different Strategy

Many problems can be solved in more than one way.

Strategies may differ in:

- how they break down the problem
- how much work they repeat
- how easy they are to explain
- how well they fit the data shape

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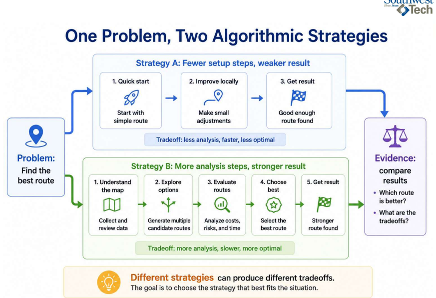

Strategy Comparison

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What Counts as Success Today

- ***Describe*** two strategies
- ***Test*** both strategies
- ***Compare*** correctness and readability
- ***Discuss*** likely growth behavior
- ***Recommend*** one strategy with conditions

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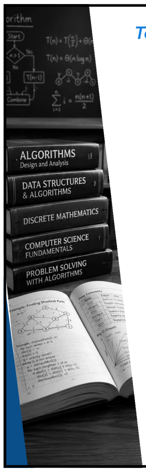

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Designing Algorithms

The reading focuses on designing algorithms.

Design means choosing a strategy that fits:

- The required result
- The problem constraints
- The data size
- The cost of the approach
- The explanation needed

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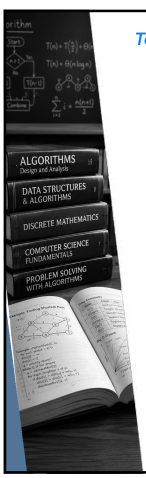

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Functional Requirements

Functional requirements ask:

- What must the algorithm produce?
- What input does it need?
- What output is expected?
- What counts as a correct result?

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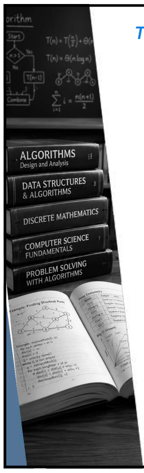

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Non-Functional Requirements

Non-functional requirements ask:

- How fast should it be?
- How readable should it be?
- How much memory can it use?
- How well should it scale?
- How explainable does it need to be?

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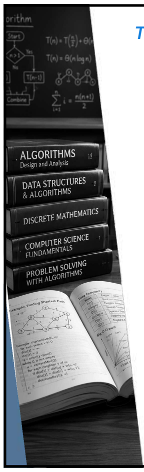

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Three Design Concerns

When designing an algorithm, ask:

1. Is it correct?
2. Is the performance reasonable?
3. Can it scale when the data or demand grows?

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Three Algorithm Design Concerns

1. Correctness	2. Performance	3. Scalability
Does it produce the expected result?	How much work does it take?	What happens as the data grows?

Three Design Concerns

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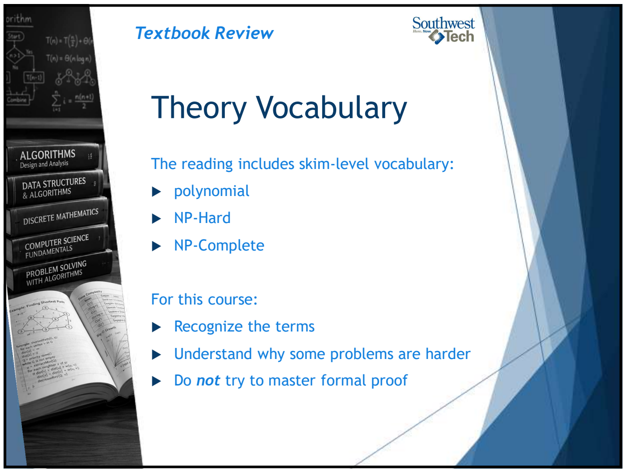
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Theory Vocabulary
The reading includes skim-level vocabulary:
polynomial
NP-Hard
NP-Complete
For this course:
Recognize the terms
Understand why some problems are harder
Do not try to master formal proof
ALGORITHMS
DATA STRUCTURES & ALGORITHMS
DISCRETE MATHEMATICS
COMPUTER SCIENCE FUNDAMENTALS
PROBLEM SOLVING WITH ALGORITHMS

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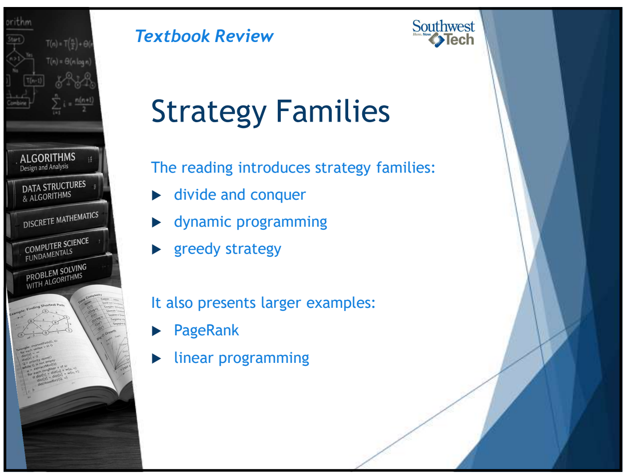
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Strategy Families
The reading introduces strategy families:
divide and conquer
dynamic programming
greedy strategy
It also presents larger examples:
PageRank
linear programming
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DATA STRUCTURES & ALGORITHMS
DISCRETE MATHEMATICS
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PROBLEM SOLVING WITH ALGORITHMS

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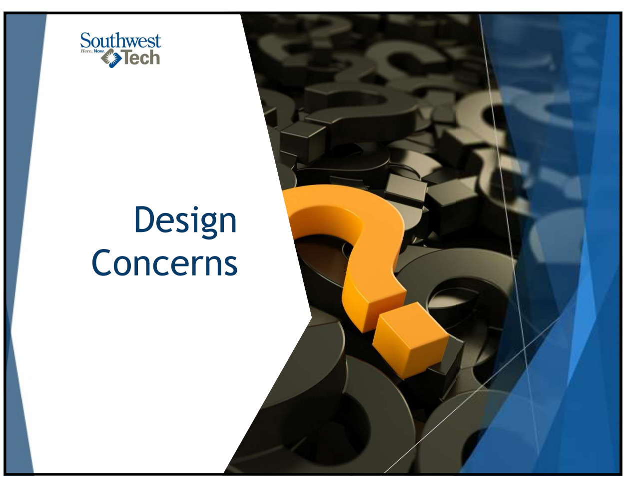
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Design Concerns

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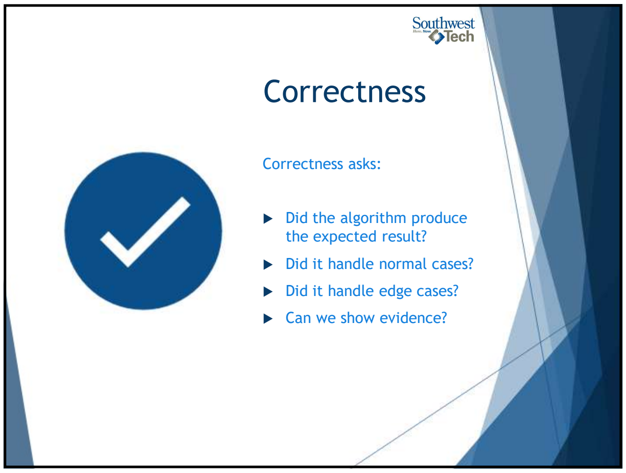
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Correctness
Correctness asks:
Did the algorithm produce the expected result?
Did it handle normal cases?
Did it handle edge cases?
Can we show evidence?

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Performance
How much work does the strategy do?
Does it repeat unnecessary work?
How does the work grow?
Is this approach reasonable for the problem size?

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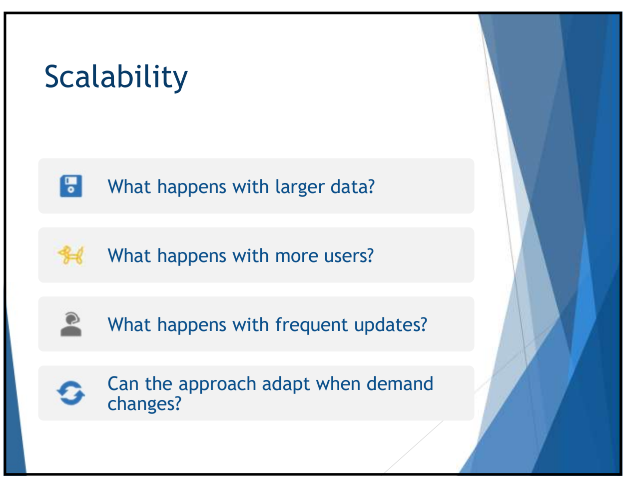
Scalability
What happens with larger data?
What happens with more users?
What happens with frequent updates?
Can the approach adapt when demand changes?

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Strategy Comparison Frame

Criterion	Strategy A	Strategy B
Correctness		
Readability		
Growth		
Fit to data		

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Better Means Better For This Context

Avoid
- "This strategy is better."

Use
- "This strategy is better for this problem because..."
- - "This strategy may not be better when..."

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Evidence For Strategy Choice

Evidence may include:
- test cases
- edge cases
- trace tables
- decision trees
- timing results
- comparison tables
- limitation notes

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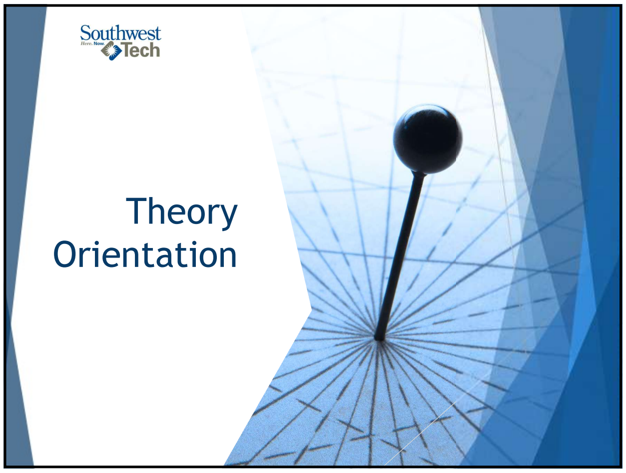
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Theory
Orientation

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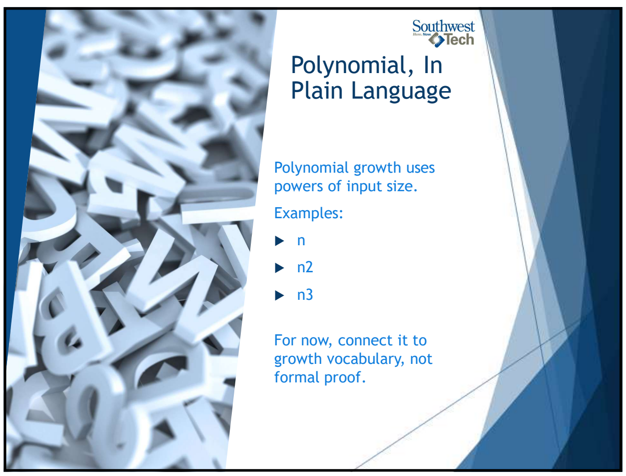
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Polynomial, In Plain Language
Polynomial growth uses powers of input size.
Examples:
n
n2
n3
For now, connect it to growth vocabulary, not formal proof.

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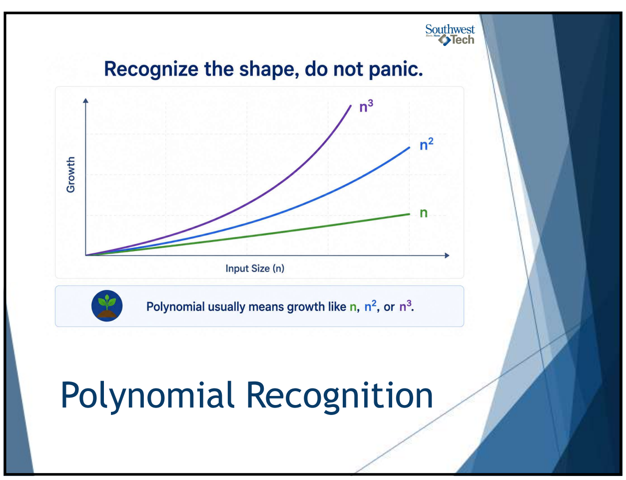
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Recognize the shape, do not panic.
Growth
Input Size (n)
n^3
n^2
n
Polynomial usually means growth like n, n^2, or n^3.
Polynomial Recognition

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NP-Hard And NP-Complete: *Recognition Only*

For this course:

- NP-Hard means very hard problem family
- NP-Complete means a special class of hard decision problems
- exact definitions require more theory

Main takeaway:

- *Some problems become hard very quickly!*

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Why This Vocabulary Matters

Advanced complexity vocabulary helps developers ask:

- Is an exact solution realistic?
- Is a good-enough solution acceptable?
- Should we limit the problem size?
- Should we use a heuristic?
- Should we ask for expert review?

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Don’t Panic Over Theory

You are not expected to master formal complexity theory today.

You are expected to:

- recognize that problem difficulty varies
- connect strategy choice to constraints
- explain tradeoffs honestly
- use AI and references carefully

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Strategy Is A Design Decision
A strategy decides how the problem will be attacked.
Examples:
try everything
split the problem
reuse earlier results
choose the best-looking next step
follow a recursive structure

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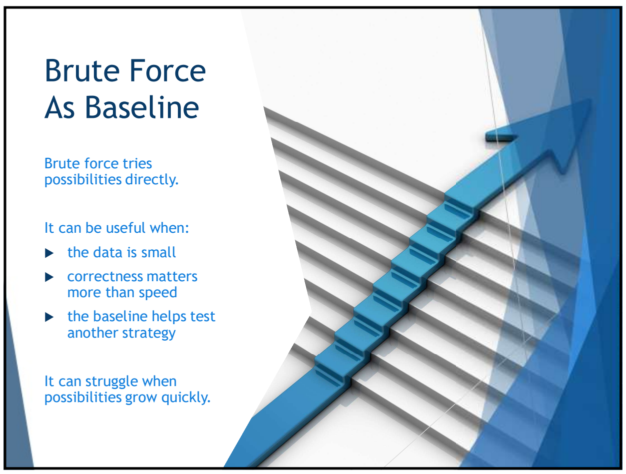
Brute Force As Baseline
Brute force tries possibilities directly.
It can be useful when:
the data is small
correctness matters more than speed
the baseline helps test another strategy
It can struggle when possibilities grow quickly.

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Break
10 Minute "Human" bio-break

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Algorithmic
Strategies

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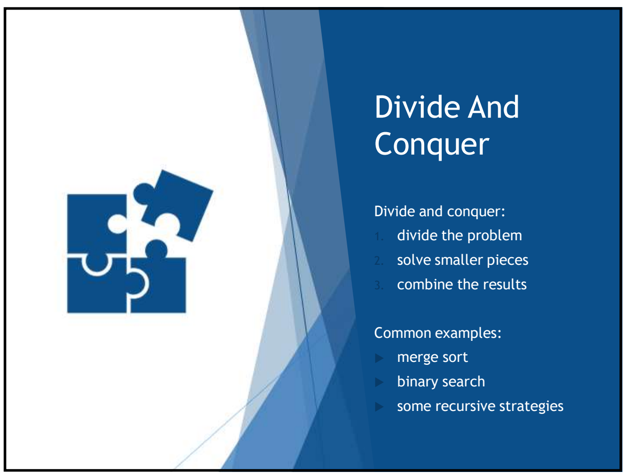
Divide And
Conquer
Divide and conquer:
1. divide the problem
2. solve smaller pieces
3. combine the results
Common examples:
merge sort
binary search
some recursive strategies

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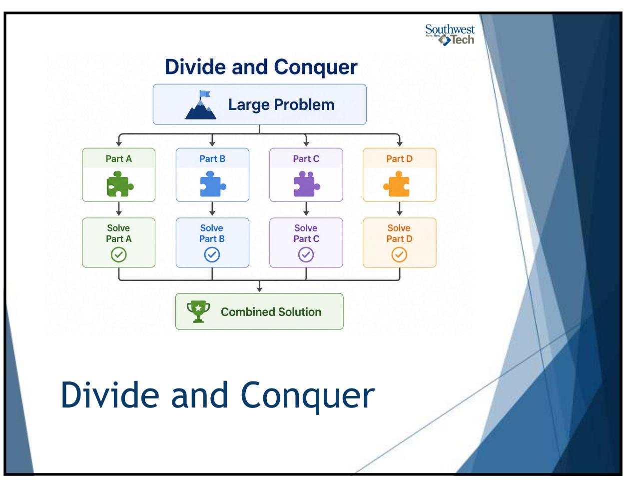
Divide and Conquer
Large Problem
Part A
Part B
Part C
Part D
Solve Part A
Solve Part B
Solve Part C
Solve Part D
Combined Solution
Divide and Conquer

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Divide And Conquer Fit

Divide and conquer fits when:

- the problem can be split
- smaller pieces are similar to the whole
- results can be combined
- splitting reduces the work or complexity

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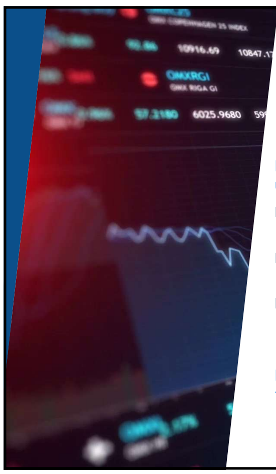

Dynamic Programming

Dynamic programming is useful when:

- the problem has repeated subproblems
- earlier results can be saved
- saved results prevent repeated work

For this course: recognize the pattern.

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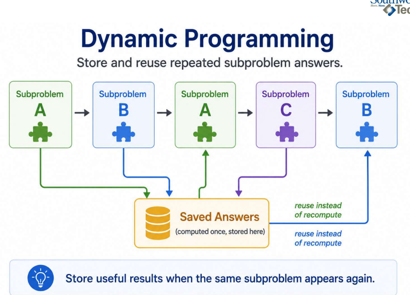

Dynamic Programming

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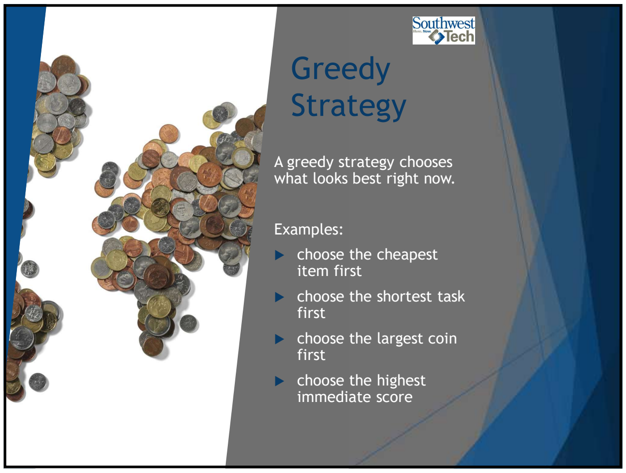
Greedy Strategy
A greedy strategy chooses what looks best right now.
Examples:
choose the cheapest item first
choose the shortest task first
choose the largest coin first
choose the highest immediate score

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Greedy Risk
Greedy strategies can fail when:
an early choice blocks a better later choice
the local score hides a global tradeoff
the rule does not match the real goal
Greedy needs evidence

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The Risk of a Greedy Algorithm
Start
Short-term choice:
Immediate reward: 8
...
...
...
Dead end / weaker result
Final total:
10
Patient choice:
Immediate reward: 4
...
...
...
...
Better later outcome
Final total:
18
Best next step is not always best overall.
Greedy Risk

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Strategy Family Comparison

Strategy	Basic Idea	Watch For
Brute force	try possibilities	too much work
Divide and conquer	split and combine	combine step
Dynamic programming	reuse saved results	identifying repeated subproblems
Greedy	best-looking next step	local choice may fail

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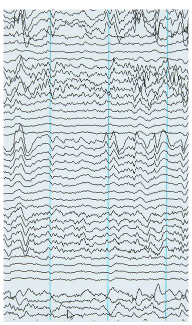

PageRank As An Example

It considers:

- links between pages
- importance passed through relationships
- repeated updating of scores
- ranking based on structure

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PageRank: What To Notice

Notice the algorithmic questions:

- What is represented?
- What is repeated?
- What score changes over time?
- What does the rank claim?
- What are the limits of that claim?

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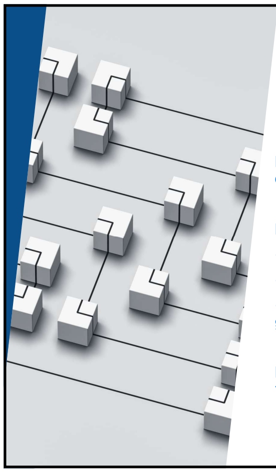

Linear Programming: *Recognition*

Linear programming helps optimize under constraints.

Plain-language frame:

- choose values
- follow constraints
- maximize or minimize a goal

For this course: recognize the idea.

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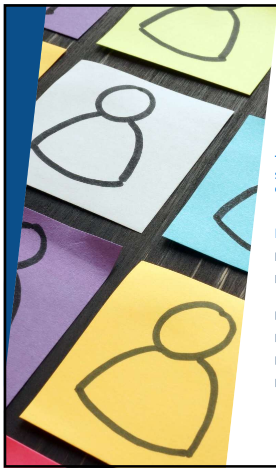

Constraints Shape Strategy

The same problem can change strategy when constraints change.

Examples:

- exact answer required
- good-enough answer acceptable
- small data
- large data
- limited memory
- explanation required

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Strategy Before Code

Before coding, describe:

- Strategy A
- Strategy B
- what each strategy assumes
- what evidence will compare them
- what would make one a better fit

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Same Strategy, Different Form

Sometimes the strategy changes.

Sometimes the representation changes.

Sometimes both solutions solve the same task, but one fits the data shape more clearly.

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Iteration and Recursion

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Iteration

Iteration repeats steps using a loop.

Common signs:

- `for`
- `while`
- running total
- index or counter
- explicit work list or stack

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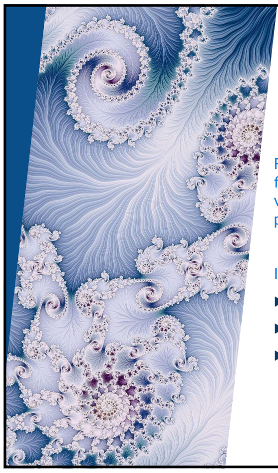

Recursion

Recursion happens when a function solves a smaller version of the same problem.

It needs:

- a base case
- a recursive step
- progress toward stopping

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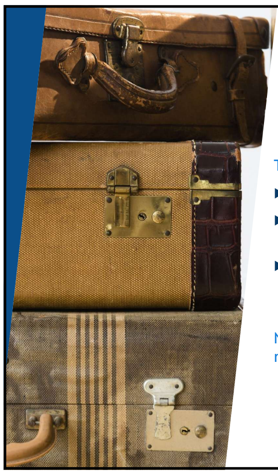

Base Case

The base case answers:

- When are we done?
- What result can we return directly?
- How do we stop the repeated calls?

No base case means no reliable recursion.

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Recursive Trace

A recursive trace can show:

- call depth
- current input
- action taken
- returned result
- how totals combine

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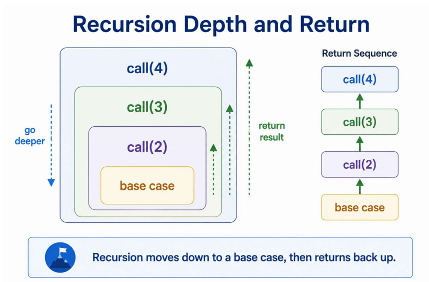

Recursive Trace

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Iteration vs Recursion

Question	Iteration	Recursion
How does it repeat?	loop	function calls itself
What must be tracked?	loop state	call state and base case
When does it fit?	flat repeated work	nested or self-similar work

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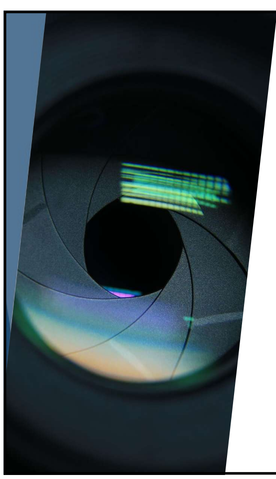

Demo

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Demo Scenario

Problem:

Compare two strategies:
- iterative processing with an explicit work stack
- recursive processing of nested groups

Goal

Calculate the total value of nested donation envelopes

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Demo Evidence

- Iterative total
- Recursive total
- Recursive call trace
- Strategy comparison table
- Key takeaway

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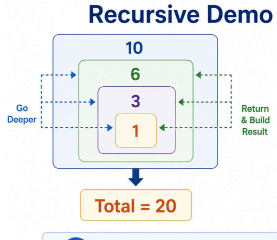
Recursive Demo Trace



Trace Table

Depth	Value	Running Total
1	10	10
2	6	16
3	3	19
4	1	20

Trace evidence makes the hidden steps visible.

Demo Evidence

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Demo Comparison

Compare:

- correctness
- readability
- growth
- fit to data

The answer alone is not the whole evaluation.

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Demo Explanation

“Both strategies produce the same total. The recursive strategy fits the nested data more directly because each nested group can be treated as a smaller version of the same problem.”

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From Demo to Lab

In Lab 05 you will compare two strategies for a different problem.

Possible options include:

- cumulative product
- grouped values
- decision tree
- coin-change or greedy selection
- scheduling or shopping tradeoffs

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Lab 05 Evidence

problem statement	two strategy descriptions	two implementations or simulations
at least four tests	trace, decision tree, or comparison table	recommendation and limitation

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README Evidence

Your README should show:

- what problem you solved
- what two strategies you compared
- what tests you ran
- what evidence you collected
- which strategy you recommend
- when your recommendation might change
- AI-use note, if applicable

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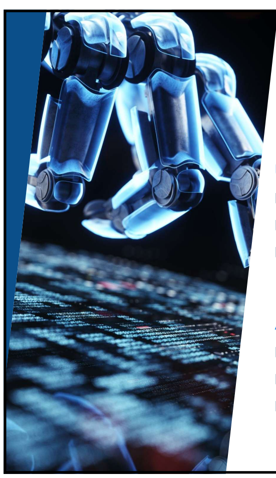

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AI Use In Lab On

Use AI after you have:

- framed the problem
- described your first strategy
- attempted or planned evidence

AI may help:

- suggest a second strategy
- critique tradeoffs
- debug one implementation

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Wrap up

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What To Carry Forward

Strategy choice is contextual.

Ask:

- Does it produce the expected result?
- Is it readable enough to explain?
- How does the work grow?
- Does it fit the data shape?
- What tradeoff did I accept?

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How To Use The Textbook For Next Week's Reading

Focus on:

- graph basics: nodes, edges, links, and networks
- graph representation
- simple, directed, undirected, and weighted graphs
- ego networks and neighborhoods
- shortest path intuition
- BFS and DFS traversal

Optional reading includes centrality, density, triangles, graph visualization, and social network analysis

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